

A shield attitude prediction method considering downtime-excavation dual states and gated attention

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Abstract: Accurate prediction of shield machine attitude is essential for safe and efficient tunnel construction. This study proposes a novel approach termed the Downtime-Excavation Attitude Model (DEAM), which integrates a downtime attitude model, an excavation model, and gated attention to capture key features under different operating states. The method was validated using data collected from the super-large-diameter shield tunnel project in Haizhu Bay, Guangzhou. For pitch angle attitude (PAA) prediction, DEAM achieved an R^2 value of 0.92, with a root mean square error (RMSE) of 0.69° and a mean absolute error (MAE) of 0.39° , outperforming existing approaches. Ablation experiments showed that geological parameters are critical, while SHAP-based feature importance analysis identified chamber pressure, hydraulic cylinder parameters, burial depth, and return slurry flow rate as dominant factors. These findings support robust attitude prediction and targeted optimization of shield attitude control.

Keywords: Shield tunnel, Shield attitude, Deep learning, Pitch angel, Gated attention

1 Introduction

With the rapid development of urban rail transit systems, the number of tunnel engineering projects has continued to increase [1]. As a tunnelling method characterized by a high degree of automation, enhanced safety, and minimal surface disturbance, shield tunnelling technology has been

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24 widely applied in the construction of urban subway and highway tunnels [2,3]. During shield
25 tunnelling construction, the shield machine is required to advance strictly along the Design Tunnel
26 Axis (DTA). However, owing to geological complexity, delayed adjustment of shield operating
27 parameters, and uneven load distribution, achieving precise guidance along the DTA remains
28 challenging [4,5]. The deviation caused by the shield machine departing from the DTA is referred to
29 as Shield Tunnelling Misalignment (STM), which is recognized as one of the major factors affecting
30 tunnel construction quality [6]. Excessive STM may result in severe disturbances to the surrounding
31 environment, segment misalignment, and other engineering problems [7]. At present, shield attitude
32 control mainly relies on feedback-based methods. After deviations occur, operators adjust operating
33 parameters according to real-time measurements obtained from the automatic guidance system.
34 However, this approach suffers from untimely control actions and delayed adjustments [8]. Therefore,
35 achieving accurate prediction of shield machine attitude, thereby providing operators with reliable
36 guidance for parameter adjustment, is of great significance for improving the quality and safety of
37 tunnel construction.

38 In recent years, shield machine attitude prediction has attracted extensive attention from
39 researchers. Traditional approaches mainly rely on physical experiments, statistical numerical
40 analysis, and mechanical principles, with emphasis on revealing the intrinsic interaction mechanisms
41 between the shield machine and surrounding ground. Based on mechanics, kinematics, and field
42 monitoring data, researchers have established interaction models between the shield machine and
43 surrounding soil to analyze its kinematic behavior and its influence on the ground. In addition, some
44 studies have attempted to reduce dependence on specific engineering conditions by simplifying
45 external loading conditions and deriving analytical solutions to the governing mechanical equations
46 [5,9]. However, shield machine attitude prediction is inherently a nonlinear problem involving
47 multiple inputs and large datasets, which limits the adaptability of traditional methods to varying
48 geological environments and makes real-time prediction difficult to achieve [10].

49 Currently, with the increasing integration of artificial intelligence across various disciplines,
50 machine learning techniques have attracted growing attention [11]. To overcome the limitations of
51 traditional methods, researchers have gradually shifted toward data-driven machine learning

algorithms. A wide range of machine learning methods have been applied to the prediction of shield tunnelling parameters, with the core objective of learning the nonlinear relationships among construction parameters, geological information, and shield machine responses from historical data. Xu et al. [12] compared several algorithms, including Bayesian ridge regression, random forest, and gradient boosting trees, to predict TBM penetration rate, cutterhead rotation speed, thrust, and torque, providing a reference for model selection in tunnelling parameter prediction. Huang et al. [13] combined genetic algorithms and principal component analysis with MLP, SVM, and GBR to establish a shield machine attitude prediction model, which was further used for the optimization and control of construction parameters. Chen et al. [14] developed a Bayesian-LGBM-based shield attitude prediction model and identified key construction parameters affecting shield attitude through feature importance analysis, thereby improving both prediction accuracy and interpretability. For the prediction of key tunnelling parameters in earth pressure balance shields, Wang et al. [15] established a predictive model based on LGBM and Bayesian optimization to estimate total thrust, cutterhead torque, cutterhead rotation speed, and advance rate, further demonstrating the applicability of ensemble learning methods in shield tunnelling parameter prediction. In addition, some studies have focused on improving feature engineering, regarding it as a crucial factor for enhancing predictive performance. Shen et al. [16] integrated wavelet denoising with a random forest model and selected the optimal combination of input parameters based on Pearson correlation coefficients, indicating that a well-designed input feature set is more conducive to improving shield attitude prediction accuracy. Zhao et al. [17] constructed physics-enhanced features from multiple dimensions, including time, historical attitude, tunnelling performance, thrust ratio, parameter variation rate, cross-combined variables, and geological characteristics, and proposed the EII index to select critical features, reducing shield attitude prediction errors. Although these machine learning approaches have achieved substantial progress, their capability to capture complex nonlinear relationships remains limited, and they still face considerable challenges when dealing with time-series data characterized by high-frequency variations.

Deep learning models have become a major research focus owing to their strong capabilities in feature abstraction and time-series modelling. Among them, recurrent neural networks and their

80 variants, represented by long short-term memory (LSTM) networks, have been widely adopted.
81 Incorporating an attention mechanism into LSTM can enhance the model's ability to capture critical
82 temporal information and improve prediction accuracy [18]. Compared with various ensemble
83 learning algorithms, LSTM is better suited for modelling long-term dependencies [19]. To further
84 improve predictive performance, an increasing number of hybrid deep learning architectures have
85 been proposed. These models typically integrate signal processing, spatial feature extraction, and
86 temporal prediction modules. Zhou et al. [6] combined wavelet denoising, CNN-based feature
87 extraction, and LSTM prediction to achieve high-accuracy multi-step prediction. The WT-Adam-
88 LSTM model proposed by Shen et al. [20] enhanced the ability of LSTM to capture spatiotemporal
89 correlations through wavelet transform. Bai et al. [10] integrated one-dimensional CNN with GRU to
90 directly process raw time-series data and demonstrated favorable robustness in multi-step prediction.
91 Li et al. [21] combined data-driven models, physical laws, and multi-task learning, and employed an
92 adaptive sequential quadratic programming algorithm to achieve high-precision inertial effect
93 matching and attitude control of tunnel boring machines. Fu et al. [22] integrated PCA with temporal
94 convolutional networks for real-time prediction of super-large-diameter shield machines. Xu et al.
95 [23] combined CNN, LSTM, and transfer learning, and verified the adaptability and stability of the
96 proposed framework under various geological conditions. Wang et al. [24] integrated deep
97 reinforcement learning, GRU, and the DDPG model to achieve dynamic TBM attitude control,
98 improving tunnelling efficiency and attitude accuracy in stable environments while exhibiting strong
99 robustness in complex underground conditions. In addition, several studies have introduced signal
100 decomposition strategies. Wang et al [25]. decomposed time-series data into trend and fluctuation
101 components and employed multi-head attention for component-wise prediction, thereby reducing the
102 risk of misleading operational decisions in attitude prediction. Advances in these predictive models
103 have ultimately promoted the development of engineering applications toward higher levels of
104 intelligence. Hu et al. [26] systematically described an intelligent control system for self-driving
105 shield tunnelling, and its application in real engineering projects demonstrated superior performance
106 in attitude control. These studies have achieved notable progress and confirmed the effectiveness of
107 deep learning in shield performance prediction tasks. Nevertheless, existing data-driven studies have

108 mostly focused on dynamic parameters during the tunnelling stage, while downtime data are often
109 neglected, leaving their potential value insufficiently exploited. Moreover, most deep learning models
110 fail to fully account for the differences in force mechanisms and attitude response patterns of shield
111 machines under different operating states when processing time-series data.

112 To address the aforementioned issues, this study proposes a novel method termed DEAM. The
113 proposed method integrates a downtime attitude model with an excavation model and introduces
114 gated attention to extract critical features from the downtime and tunnelling states, respectively,
115 thereby enabling high-accuracy prediction of the PAA. The downtime attitude model uses monitoring
116 data collected during downtime to capture the mechanical relationship between the shield machine
117 and the surrounding ground under equilibrium conditions, whereas the excavation model corrects the
118 static prediction results based on dynamic data from the tunnelling stage. The gated attention
119 mechanism enhances the model's ability to adaptively filter noise and emphasize critical features,
120 thereby improving both prediction performance and robustness.

121 **2 Methodology**

122 To efficiently and accurately predict shield machine attitude, this study proposes a novel method
123 termed DEAM. The proposed method integrates gated attention with a decomposed framework based
124 on static soil-shield interaction, extracting critical features from both downtime and tunnelling states
125 to enhance the prediction of attitude variations under complex construction conditions. The downtime
126 attitude model uses monitoring data collected during shield downtime to establish the relationship
127 between mechanical loading and attitude, thereby capturing the response patterns of the system under
128 equilibrium conditions. By calculating the residual between the measured PAA and the value
129 predicted by the downtime attitude model, the excavation model further incorporates tunnelling-stage
130 parameters as inputs to correct the static prediction results. The gated attention enhances the
131 nonlinearity and sparsity of the attention layer, thereby improving training stability and prediction
132 accuracy. The overall framework of the developed method is illustrated in Fig. 1. Each component,
133 including data preparation, deep learning network development, performance evaluation, and model
134 interpretation, is described in detail in the following sections.

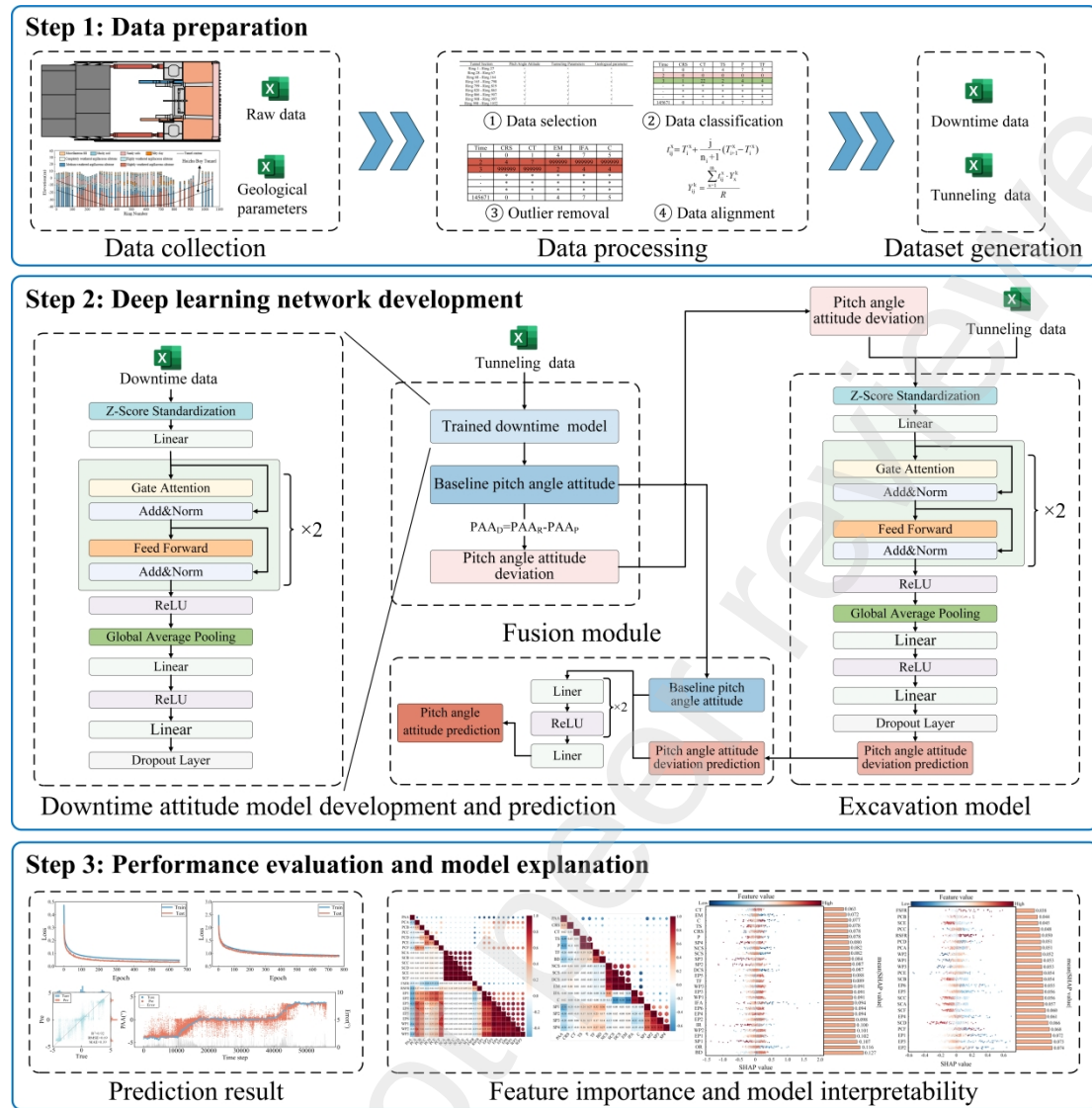


Fig. 1 Flowchart of the developed method.

2.1 Data preparation

Data processing was performed directly on the raw data collected from the shield construction data management system, and the quality of these data directly determines the upper bound of model performance [27]. In this study, data processing mainly involved data selection, data classification, outlier removal, data alignment, and parameter standardization. These procedures were designed to improve the regularity and consistency of the dataset, thereby enhancing the training efficiency and testing accuracy of the gated-attention-based deep learning model.

2.1.1 Data selection

This study is based on the eastbound line of the Haizhu Bay Tunnel project, which comprises approximately 1,000 segment rings. Table 1 presents a statistical summary of the dataset. To ensure

the completeness and reliability of the research data, ring sections with incomplete geological records or missing key tunnelling parameters were excluded. After screening, construction data corresponding to approximately 700 segment rings were ultimately adopted as the research dataset. The dataset contains more than 680,000 real-time monitoring and tunnelling parameter records, providing sufficient data support for subsequent in-depth analysis.

Table 1 Data availability of the shield tunnelling dataset.

Tunnel Section	PAA	Tunnelling Parameters	Geological parameters
Ring 1 - 27	×	×	×
Ring 28 - 67	√	×	×
Ring 68 - 164	√	√	×
Ring 165 - 798	√	√	√
Ring 799 - 819	√	√	×
Ring 820 - 865	√	√	√
Ring 866 - 907	√	√	×
Ring 908 - 997	√	√	√
Ring 998 - 1052	√	√	×

Note: A tick (√) indicates available data, whereas a cross (×) indicates missing data.

2.1.2 Data classification and outlier removal

During tunnel excavation, the shield machine alternates between tunnelling and downtime states. During downtime, the shield machine ceases forward advancement because of operations such as segment assembly. The data recorded by various sensors at this stage can therefore more clearly reflect the mechanical relationship between the shield machine and the surrounding ground under static conditions. In addition, as this study focuses on the accurate prediction of shield attitude during tunnelling, the raw monitoring data were first classified into downtime and tunnelling states according to the operating condition to enable more refined modelling. This classification provides the basis for state-specific modelling, allowing the model to more accurately extract the intrinsic relationships between tunnelling parameters and attitude variations under different operating states, thereby improving both the reliability and interpretability of the prediction model.

This process was implemented using Eq. (1) and (2). For the raw tunnelling construction parameters, all parameter values across different feature dimensions were traversed. Data satisfying Eq. (1) were classified as the tunnelling state, whereas data satisfying Eq. (2) were classified as the downtime state. To further improve data quality, outlier removal was performed separately for the data in the two states. Specifically, each cell was examined, and when the value in a given cell satisfied Eq. (3), the row containing the outlier was removed.

$$R_E = F_1(R) \quad (2)$$

$$R_D = F_2(R) \quad (3)$$

$$R' = F(R) \quad (4)$$

where R_E and R_D denote the classified tunnelling dataset and downtime dataset, respectively, while $F_1(R)$ and $F_2(R)$ represent the corresponding operations. $F_1(R)$ indicates that when the product of tunnelling parameters, including cutter rotational speed, cutter torque, thrust speed, penetration, and thrust force, is nonzero, the corresponding data group is classified as tunnelling data. In contrast, $F_2(R)$ indicates that when all tunnelling parameters are equal to zero, the corresponding data group is classified as downtime data. R' denotes the dataset after outlier removal. $F(R)$ indicates that when the value in a cell exceeds 10 times the mean value of its corresponding feature dimension, the row containing that cell is removed.

2.1.3 Data alignment

All types of data should be structured to satisfy the input requirements of machine learning algorithms [13]. Since the number of soil layers intersected by each tunnel cross-section varies, the soil properties at each cross-section were weighted and aggregated to ensure the consistency of input data. For each soil layer, six parameters closely related to tunnelling construction and mechanical behavior were extracted from the site investigation report. The soil parameters for each geological stratum were assigned using the average values provided in the investigation report. Consequently, variations in the soil properties of tunnel cross-sections were primarily caused by changes in the thickness of different strata. Because the information provided by boreholes was limited, the soil properties could not be directly matched to each row of tunnelling data. To associate geological information with tunnelling data, linear interpolation was employed to estimate the thickness of soil layers between adjacent boreholes, and the interpolated values were subsequently mapped to each collected tunnelling data sample. Specifically, Eq. (4) was used to calculate the thickness of each soil layer for every sample, and Eq. (5) was then applied to determine the corresponding cross-sectional soil parameters. This processing approach aligns discrete and stratified geological investigation data with high-frequency tunnelling data, thereby ensuring the consistency and effectiveness of the model

198 inputs.

$$199 \quad t_{ij}^x = T_i^x + \frac{j}{n_i + 1} (T_{i+1}^x - T_i^x) \quad (5)$$

$$200 \quad V_{ij}^k = \frac{\sum_{x=1}^m t_{ij}^x \cdot Y_x^k}{D} \quad (6)$$

201 where t_{ij}^x denotes the thickness of the x -th soil layer corresponding to the j -th sample in the i -th
202 interval; T_i^x and T_{i+1}^x represent the thicknesses of the x -th soil layer in the i -th and $(i+1)$ -th boreholes,
203 respectively; n_i denotes the number of samples within the i -th interval; and V_{ij}^k represents the value
204 of the k -th geological parameter corresponding to the j -th sample in the i -th interval. Furthermore, m
205 denotes the total number of soil layers intersected by the tunnel cross-section, Y_x^k represents the
206 average value of the k -th geological parameter for the x -th soil layer provided in the geotechnical
207 investigation report, and D denotes the shield diameter.

208 This study focuses on the eastbound tunnel data. Fig. 2 illustrates the transverse soil profile
209 along the tunnel alignment, where the thickness of soil strata varies with shield advancement. The
210 geological formations traversed by the shield machine mainly consist of Completely Weathered
211 Argillaceous Siltstone, Highly Weathered Argillaceous Siltstone, Medium Weathered Argillaceous
212 Siltstone, and Slightly Weathered Argillaceous Siltstone. In the middle section of the tunnel, the
213 shield machine passes through extensive layers of Medium Weathered Argillaceous Siltstone and
214 Slightly Weathered Argillaceous Siltstone. The tunnel is characterized by a composite soft-hard
215 ground condition, with soft soil in the upper part and hard rock in the lower part. Under such
216 geological conditions, the shield machine is prone to uneven loading during tunnelling, resulting in
217 deviations of the actual tunnelling axis from the designed alignment. These challenges make shield
218 attitude control one of the most critical aspects of tunnel construction.

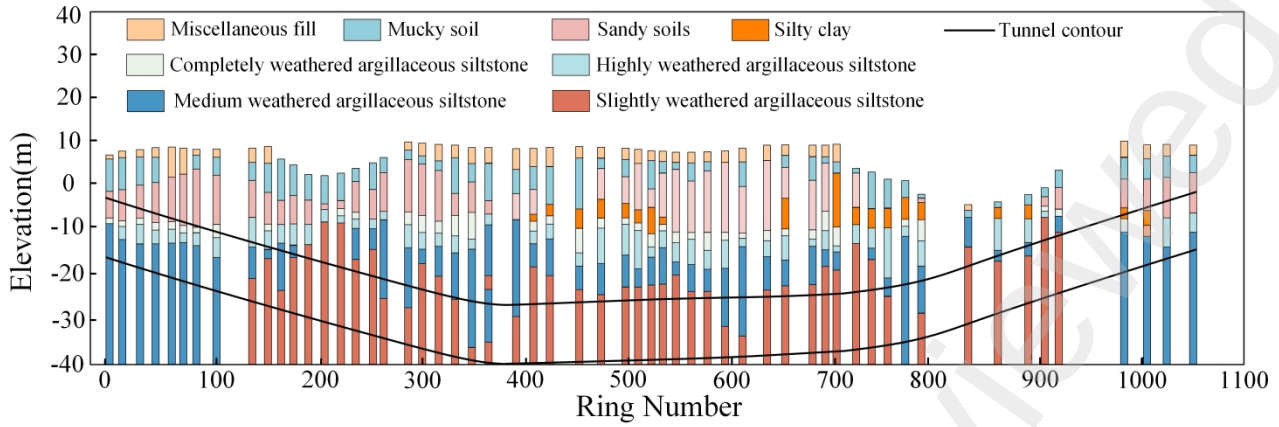


Fig. 2 Soil layers in the boreholes for the east tunnel line.

2.1.4 Data normalization

Data normalization aims to eliminate feature dominance caused by differences in units and scales by scaling or transforming raw features into a unified range, thereby ensuring that each feature contributes equally during model training [28]. By promoting balanced consideration of all features, data normalization improves model accuracy, stability, and convergence speed. The input parameters include six geological parameters, tunnel burial depth, and 33 shield operating parameters. Since these parameters are not expressed in the same dimensional scale, it is necessary to standardize all input variables to reduce the influence of dimensional differences on model convergence. In this study, Z-score normalization was adopted. Specifically, for a given dataset R , the values of different feature dimensions were statistically analyzed to obtain the mean μ and standard deviation σ for each feature dimension. Based on these statistics, the normalized value was calculated using Eq. (6).

$$Z = \frac{R - \mu}{\sigma} \quad (7)$$

2.2 Gated attention

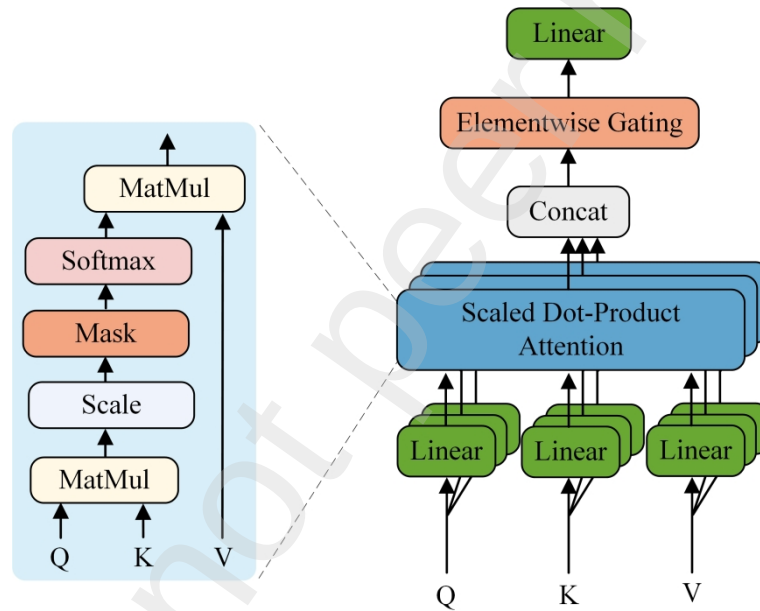
The gated attention was proposed by Qiu et al. in 2025 [29] to alleviate the limitations of the standard Transformer attention mechanism, including restricted representation capability, training instability, and unreasonable Softmax weight allocation. Its core concept is to regulate information flow through dynamic gating, thereby enhancing the expressive capacity and stability of the model.

As illustrated in Fig. 3, the gated attention introduces a learnable gating unit after each attention head based on the classical multi-head attention architecture. Specifically, the input sequence is first linearly projected, and the preliminary attention output is obtained through scaled dot-product

241 attention. Subsequently, this output is passed through a Sigmoid activation function to generate a
 242 gating vector with values ranging from 0 to 1. The gating vector is then multiplied element-wise with
 243 the attention output, enabling fine-grained modulation of features across different dimensions. The
 244 corresponding formulation is given in Eq. (7).

$$245 \quad Y' = g(Y, X, W_\theta, \phi) = Y \odot \phi(XW_\theta) \quad (8)$$

246 where Y denotes the input to be modulated, g represents the gating function, X is another input
 247 used to compute the gating scores, W_θ denotes the learnable parameter matrix of the gate, $\odot$
 248 represents element-wise multiplication, ϕ is the activation function, and Y' denotes the output after
 249 gated modulation.



250
 251 Fig. 3 Gated attention.

252 2.3 Deep learning network development

253 During shield tunnelling, the shield machine is subjected to various types of loads, mainly
 254 including self-weight, hydraulic jack thrust, cutterhead cutting loads, and interaction forces between
 255 the shield shell and the surrounding ground. Among these factors, the ground reaction acting on the
 256 shield shell is strongly influenced by nonlinear ground responses and dynamic soil-shield interactions,
 257 making it difficult to predict accurately prior to construction. As operating conditions continuously
 258 change, multiple factors, such as tunnelling parameters and chamber pressure, fluctuate throughout
 259 the excavation process, causing the soil-shield system to gradually transition from a quasi-static state

260 to a transient response state. This significantly increases the difficulty of establishing an accurate
261 theoretical calculation model for shield tunnelling attitude prediction. Moreover, the nonlinear
262 characteristics and complex construction conditions commonly encountered in practical engineering
263 often lead to substantial discrepancies between the predictions of physics-based models relying on
264 empirical constants and the measured data.

265 Notably, existing data-driven studies have mainly focused on dynamic data during the tunnelling
266 stage, whereas data corresponding to static operating conditions, such as downtime and segment
267 installation, are often neglected because of their relatively small variations or their differences from
268 tunnelling mechanisms. As illustrated in Fig. 4, under downtime conditions, the loads acting on the
269 shield machine mainly include self-weight, soil friction, surrounding earth and water pressures, and
270 hydraulic jack thrust, resulting in more evident attitude response patterns. From the perspective of
271 force mechanisms, the soil-shield system during downtime can be approximately regarded as being
272 in an equilibrium state, where the load composition is relatively explicit. In addition, static data,
273 including those collected during downtime and segment installation stages, account for a considerable
274 proportion of engineering practice and can reflect the relationship between loading conditions and
275 shield attitude under equilibrium states. Therefore, the proposed framework incorporates static-
276 condition data to characterize the correspondence between shield loading and attitude under
277 equilibrium conditions. On this basis, deviations induced by dynamic effects during the tunnelling
278 stage are further analyzed, thereby improving the accuracy of shield attitude prediction.

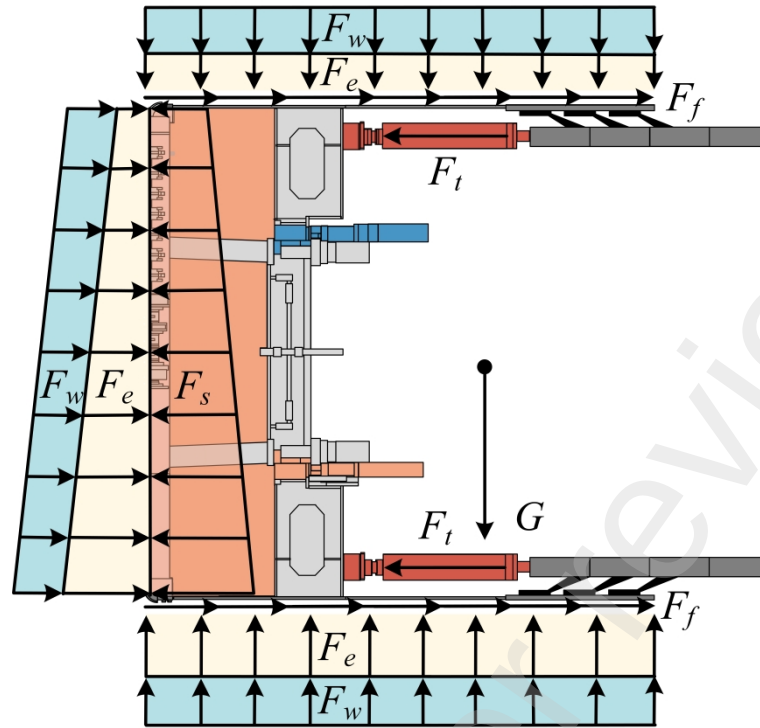


Fig. 4 Force analysis schematic of a slurry tunnel boring machine (slurry TBM) under downtime state.

Fig. 5 illustrates the workflow for PAA prediction. Specifically, data were collected from the shield construction data management system and processed to generate two datasets. One dataset consists of static condition parameters, represented by hydraulic cylinder stroke and chamber pressure, while the other is a dynamic dataset containing 21 parameters, including tunnelling parameters, geological parameters, and burial depth. First, the static parameters were used to train a downtime attitude model composed of two gated attention encoder layers, which was employed to predict the baseline value of the PAA under dynamic operating conditions. The residual was then obtained by calculating the difference between the measured value and the predicted baseline value. Subsequently, the excavation model employed another gated attention encoder with the same architecture, trained using the dynamic parameters, to predict the obtained residual. Finally, a multilayer perceptron (MLP)-based fusion module was adopted to perform weighted fusion of the baseline prediction and the residual prediction, thereby producing the final PAA prediction.

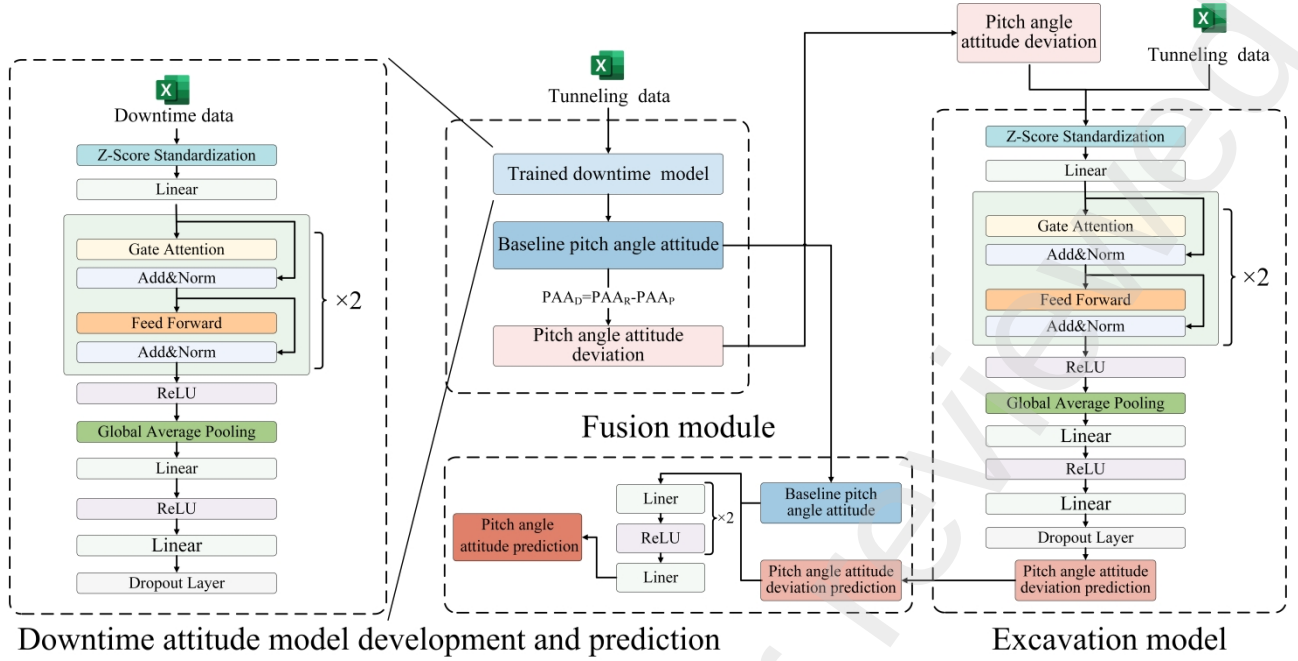


Fig. 5 General structure of DEAM.

2.4 Loss function and evaluation metrics

To continuously monitor model performance during training, the mean squared error (MSE) was adopted as the loss function in this study. This choice is justified by the fact that squaring the prediction error assigns greater penalty weights to larger errors, thereby guiding the model to prioritize the reduction of significant deviations during training. This property not only optimizes the direction of gradient updates but also enables the model to maintain stronger robustness against data fluctuations, ultimately improving overall prediction accuracy. The loss function can be expressed as follows:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (9)$$

where n denotes the total number of samples, y_i represents the true value of the i -th sample, and $\hat{y}_i$ denotes the predicted value of the i -th sample.

To evaluate the prediction accuracy of the proposed method, the MAE, RMSE, and R^2 were adopted as evaluation metrics. MAE characterizes the average absolute difference between prediction values and actual values, providing an intuitive indicator of the overall prediction deviation. RMSE quantifies the mean squared discrepancy between predictions and observations, where a smaller value indicates higher prediction accuracy. The R^2 metric reflects the fitting capability of the model, with

values closer to 1 indicating better fitting performance. By combining these evaluation metrics, the predictive performance of the model can be comprehensively assessed, enabling an in-depth analysis of its behavior under different conditions. The formulas are as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (10)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (11)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (12)$$

where n denotes the total number of samples, y_i represents the true value of the i -th sample, $\hat{y}_i$ denotes the predicted value of the i -th sample, and $\bar{y}$ represents the arithmetic mean of all true values.

3. Case study

3.1. Project overview

Model was validated using data from the Guangzhou Haizhu Bay Tunnel project. The total length of the construction alignment is approximately 4.348 km, including a tunnel section of about 3.474 km and a shield-driven section of approximately 2.078 km. The geographical location of the project is illustrated in Fig. 6. The shield tunnel project consists of a twin-bore single-deck tunnel, including the eastbound and westbound lines. The eastbound line is approximately 2,102 m long, whereas the westbound line is approximately 2.093 km long. The tunnel crown burial depth ranges from approximately 9.1 m to 36.14 m. The tunnel was excavated using a slurry TBM with a cutterhead diameter of 15.07 m. The partitioning of the shield machine hydraulic cylinders is shown in Fig. 7, and the key parameters of the shield machine are summarized in Table 2. The outer and inner diameters of the tunnel segments are 14.5 m and 13.3 m, respectively, with a segment ring width of 2 m.

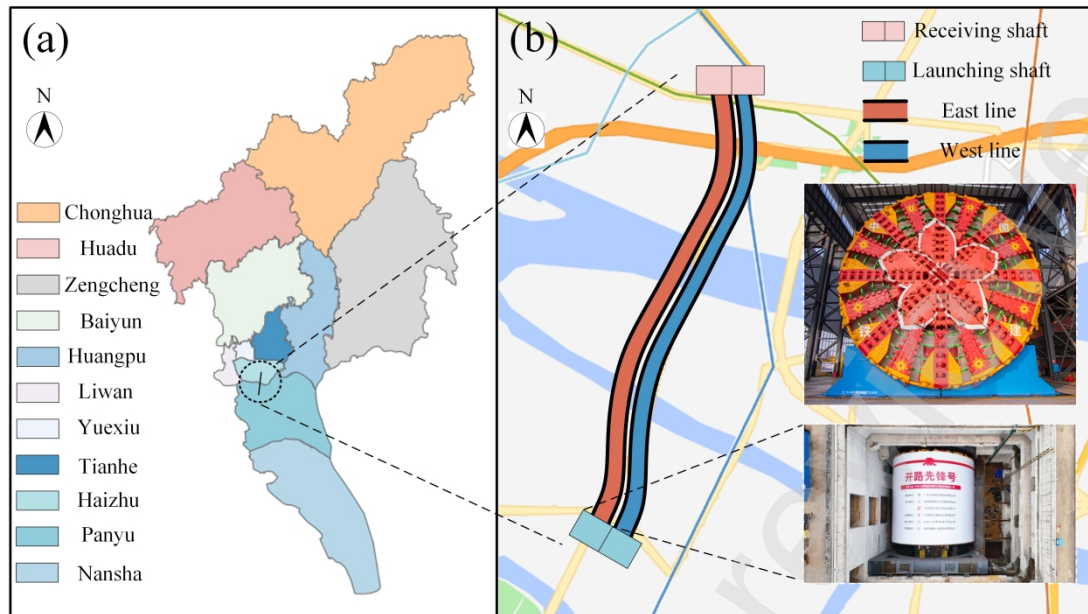


Fig. 6 Tunnel alignment drawing.

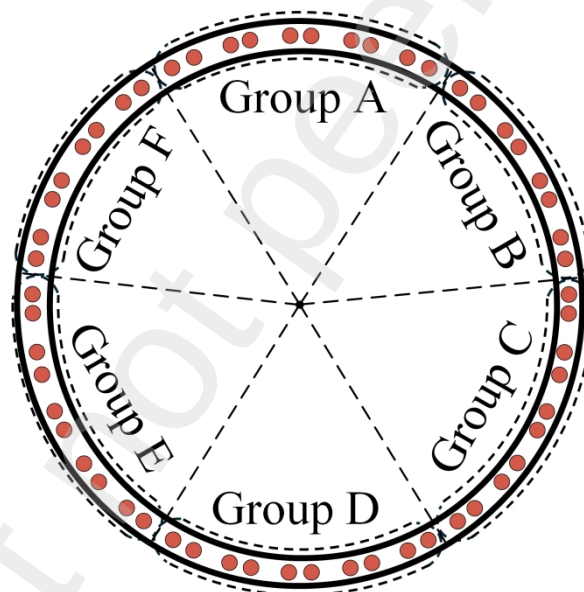


Fig. 7 Layout of the thrust jack groups

Table 2 Key parameters of the slurry TBM.

Parameters	Value
Overall Length	130 m
Maximum Advance Rate	50 mm/min
Maximum Thrust	222,173 kN 350 bar
Excavation Diameter	15070 mm
Opening Ratio	35 %

This study aims to predict the tunnelling attitude of the shield machine. Fig. 8 illustrates the definition of the PAA. The PAA is defined as the angle between the shield machine axis and the

339 horizontal plane, and it is an important parameter for describing the vertical attitude of the shield
340 machine. Therefore, it was selected as the output variable in this study. The PAA data were obtained
341 from the VMT laser guidance system.

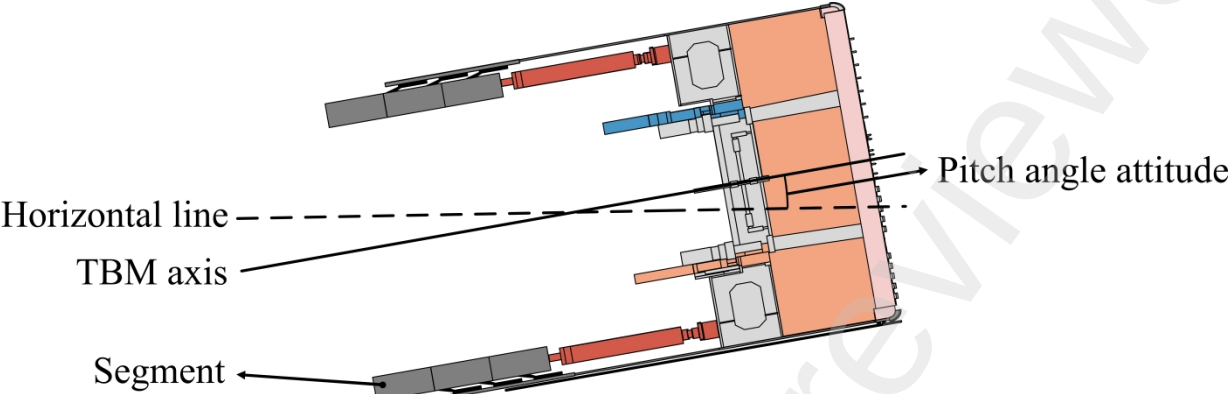


Fig. 8 Schematic diagram of the PAA.

344 As described in [Section 2.3](#) Deep learning network development, during the downtime stage, the
345 shield machine attitude is mainly governed by the combined effects of the hydraulic thrust system
346 and the pressure exerted by the surrounding ground. Among these factors, hydraulic cylinder stroke
347 and thrust pressure can directly reflect the magnitude of hydraulic thrust, while chamber pressure and
348 working chamber pressure partially characterize the static loading conditions of the surrounding
349 ground. In addition, for a slurry TBM, the feed slurry flow rate and return slurry flow rate, as key
350 parameters of the slurry circulation system, indirectly reflect the stability of the excavation face.
351 Therefore, hydraulic cylinder stroke, thrust pressure, chamber pressure, working chamber pressure,
352 feed slurry flow rate and return slurry flow rate were selected as the input variables of the static
353 attitude prediction model. During the tunnelling stage, dynamic construction parameters introduce
354 significant uncertainty into shield machine attitude. To account for these dynamic effects, chamber
355 pressure, working chamber pressure, tunnelling parameters, geological parameters, and tunnel burial
356 depth were selected as the input variables of the excavation model to correct the deviations of the
357 downtime attitude model under dynamic operating conditions. [Table 3](#) summarizes these variables.
358 All parameters were collected by the construction data management system at a sampling frequency
359 of 0.05 Hz.

Table 3 Output and input parameters.

Type	Parameter	Symbol	Unit	Min	Mean	Max
Output	Pitch angle attitude	PAA	°	-4.50	0.59	3.98

Type	Parameter	Symbol	Unit	Min	Mean	Max
Input	Pressure of cylinder group A	PCA	bar	4.20	73.56	177.34
	Pressure of cylinder group B	PCB	bar	10.52	94.33	300.92
	Pressure of cylinder group C	PCC	bar	4.04	139.59	209.30
	Pressure of cylinder group D	PCD	bar	9.34	188.31	308.77
	Pressure of cylinder group E	PCE	bar	5.67	134.82	223.59
	Pressure of cylinder group F	PCF	bar	3.02	92.89	289.36
	Stroke of cylinder group A	SCA	mm	490.53	1660.33	2865.78
	Stroke of cylinder group B	SCB	mm	513.36	1665.72	2883.25
	Stroke of cylinder group C	SCC	mm	15.42	1697.53	2916.92
	Stroke of cylinder group D	SCD	mm	593.02	1728.05	2905.77
	Stroke of cylinder group E	SCE	mm	-88.00	1709.97	2857.02
	Stroke of cylinder group F	SCF	mm	-609.54	1677.22	2878.50
	Feed slurry flow rate	FSFR	m ³ /h	1.00	2211.29	3409.00
	Return slurry flow rate	RSFR	m ³ /h	1.00	2780.82	3406.00
	Excavation chamber pressure 1	EP1	bar	1.16	2.76	4.62
	Excavation chamber pressure 2	EP2	bar	1.13	2.73	4.59
	Excavation chamber pressure 3	EP3	bar	1.31	3.06	5.16
	Excavation chamber pressure 4	EP4	bar	1.28	2.90	4.76
	Excavation chamber pressure 5	EP5	bar	2.22	3.86	5.92
	Excavation chamber pressure 6	EP6	bar	2.32	3.71	14.07
	Working chamber pressure 1	WP1	bar	0.83	3.26	4.17
	Working chamber pressure 2	WP2	bar	0.76	3.41	5.15
	Working chamber pressure 3	WP3	bar	1.67	3.27	4.18
	Cutter rotational speed	CRS	rpm	0.2	1.12	1.32
	Cutter torque	CT	MN·m	-21.22	-0.47	19.01
	Thrust speed	TS	mm/min	0.02	12.79	19.52
	Penetration	P	mm/r	0.02	10.67	30.37
	Thrust force	TF	kN	0	75655.12	93042.00
	Burial depth	BD	m	11.01	28.07	36.09
	Natural uniaxial compressive strength	NCS	MPa	20.29	21.82	22.19
	Saturated uniaxial compressive strength	SCS	MPa	10.66	18.03	19.56
	Dried uniaxial compressive strength	DCS	MPa	31.48	35.70	36.60
	Elastic modulus	EM	MPa	21300.00	21383.33	21400.00
	Internal friction angle	IFA	°	40.09	41.46	42.55
	Cohesion	C	kPa	0.22	1.72	3.31
	Shield tail grease pressure 1	SP1	bar	19.60	162.85	292.28
	Shield tail grease pressure 2	SP2	bar	35.33	247.90	403.55
	Shield tail grease pressure 3	SP3	bar	39.84	249.93	445.90
	Shield tail grease pressure 4	SP4	bar	26.76	229.83	392.26

To analyze the relationships among input parameters under different shield operating states and their influence on the PAA, Pearson correlation coefficients were used to plot parameter correlation heatmaps for the downtime and tunnelling states, respectively. The magnitude and sign of the correlation coefficient indicate the strength and direction of the linear relationship between variables [30]. Fig. 9 presents the correlation coefficient matrix between the PAA and the input parameters of the downtime attitude model under the downtime state. It can be observed that, during downtime, the correlation coefficients among hydraulic cylinder strokes are relatively high, ranging from 0.45 to 0.82. A strong correlation is also observed between chamber pressure and working chamber pressure,

369 with the maximum value reaching 1.00, indicating strong linear relationships among these input
370 parameters. Apart from the high correlations among parameters within the same system, most input
371 parameters exhibit moderate to weak correlations with one another, suggesting that parameters from
372 different systems remain relatively independent under static downtime conditions. Notably, the
373 correlations between the PAA and most input parameters are weak, with absolute values below 0.5.
374 Among them, the PAA shows the strongest positive correlation with the thrust force of hydraulic
375 cylinder group F and the strongest negative correlation with the stroke of hydraulic cylinder group E.

376 Fig. 10 shows the correlations between the input parameters of the downtime attitude model and
377 the output parameter during shield tunnelling. In the tunnelling data, the correlations among
378 parameters differ markedly from those observed in the downtime data. Extensive correlation
379 coefficients of 1.00 are observed among the strokes of different hydraulic cylinder groups, and
380 chamber pressure and working chamber pressure also exhibit a strong linear relationship. This
381 indicates that the variations in input parameters within the same system are highly consistent during
382 tunnelling. The relationships between the PAA and the input parameters also vary under different
383 operating states. Compared with the downtime state, the correlations between the PAA and hydraulic
384 cylinder strokes become weaker, whereas the negative correlations of chamber pressure and working
385 chamber pressure with the PAA are enhanced. Fig. 11 presents the correlations between input
386 parameters, including geological parameters, and the PAA. Strong correlations are observed among
387 soil parameters and shield tail grease pressure, with some correlation coefficients reaching as high as
388 0.99. Among all variables, cutterhead rotational speed, penetration, and shield tail grease pressure
389 exhibit relatively high correlations with the PAA.

390 Among the correlation coefficients between the PAA and the other input parameters of the
391 excavation model, only a few tunnelling parameters exhibit correlation coefficients exceeding 0.51,
392 while the remaining coefficients are all below 0.5. This indicates that no simple linear relationship
393 exists between the PAA and the input parameters, thereby demonstrating the suitability of using an
394 encoder-based model to capture the underlying complex nonlinear relationships. In addition,
395 significant differences are observed between the correlation structures of static and dynamic data.
396 More strong linear relationships appear in the dynamic data, and, more importantly, the key

parameters most strongly correlated with the output differ between the two operating states. This further highlights the necessity of distinguishing static data from dynamic data in the modelling process. These findings suggest that understanding the strength and direction of these correlations is crucial for interpreting how the input parameters influence the output variables in the proposed framework.

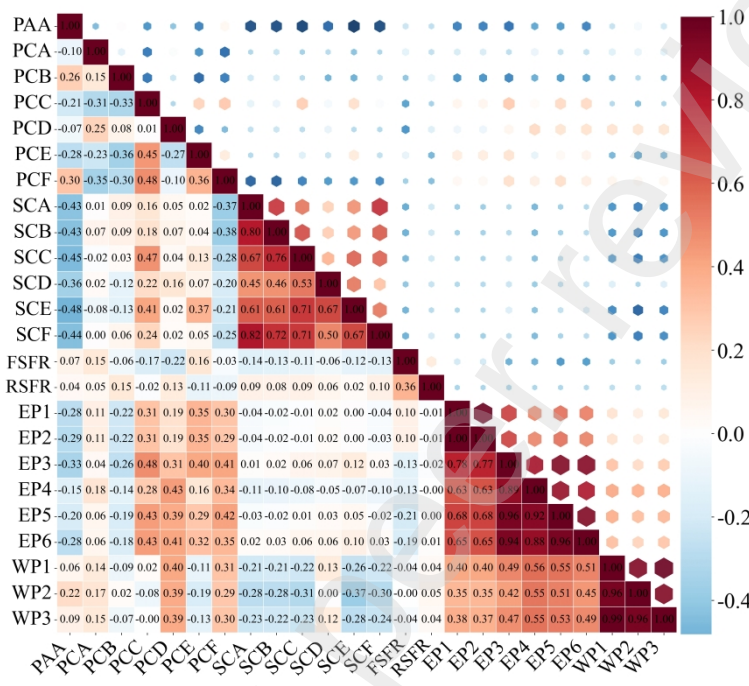


Fig. 9 Correlation of PAA and downtime attitude model inputs (downtime).

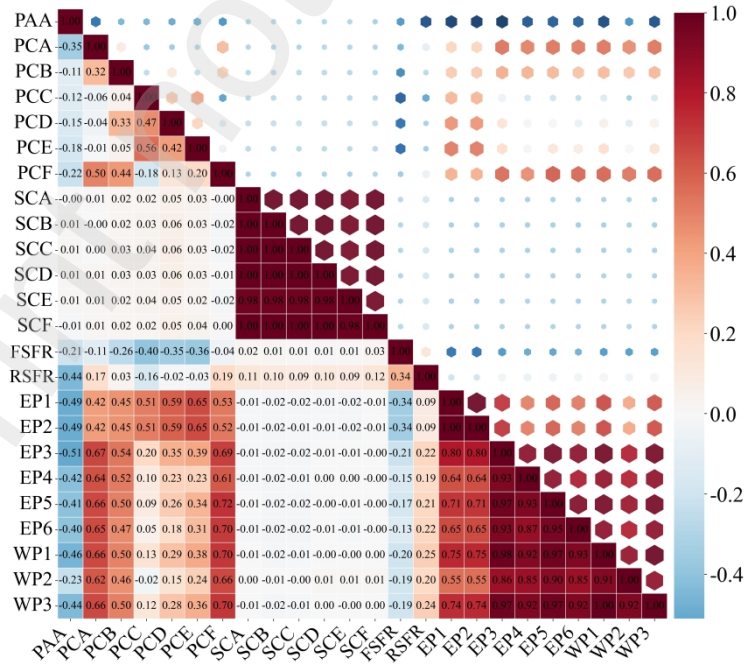


Fig. 10 Correlation of PAA and downtime attitude model inputs (tunnelling).

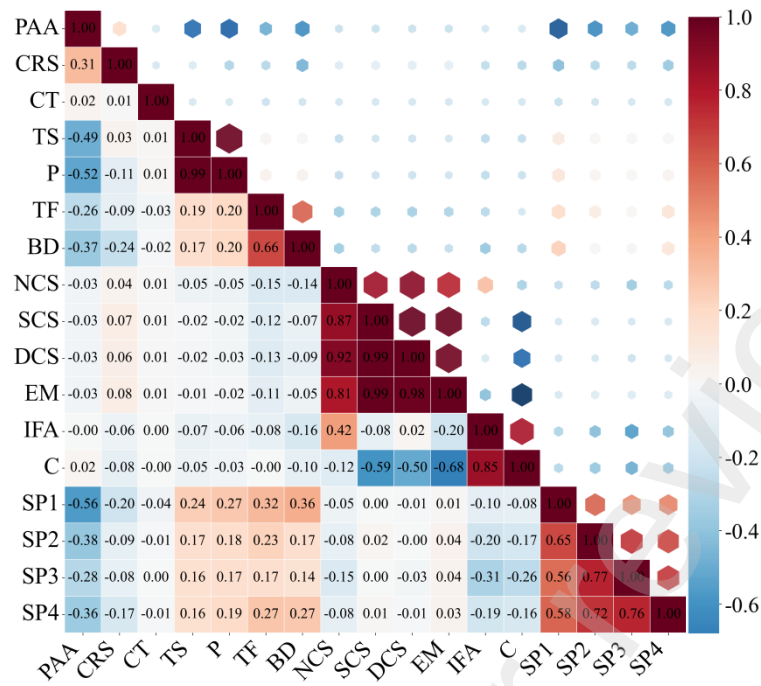


Fig. 11 Correlation of PAA and other excavation model inputs.

3.2. Prediction of the DEAM

Model training and testing were implemented using the PyTorch framework. The experiments were conducted on a computer running Windows 10 (64-bit), equipped with an AMD EPYC 7773X CPU, an NVIDIA GeForce RTX 5070 Ti GPU, and 256 GB of memory. Anaconda was used as the integrated development environment. The downtime attitude model and the excavation model adopted identical architectures and parameter settings. The dataset was divided into training, validation, and testing sets at a ratio of 6:2:2. The training parameters are listed in Table 4.

Table 4 Training parameters of DEAM.

Parameters	Value
Learning rate	0.001
Batch size	1024
Epochs	1000
Patience	50

3.2.1 Prediction results

The proposed DEAM method was evaluated on the divided dataset and compared with a gated attention Transformer model trained using only tunnelling data. The corresponding evaluation metrics were employed to assess model performance. First, the training process was recorded and monitored to observe and evaluate model behavior, thereby ensuring that a well-trained model was obtained. Subsequently, the trained model was applied to the testing set to further investigate its predictive performance.

Fig. 12 shows the training processes of the downtime attitude model and the excavation model in the prediction task. For the downtime attitude model, the loss value decreases rapidly during the first 50 epochs. When the number of training epochs exceeds 100, the loss function gradually converges to its minimum value, with only slight local fluctuations, indicating that 200 epochs are sufficient to complete the training process. For the excavation model, the training process becomes sufficiently complete after 400 epochs. Fig. 13 presents the training process of the gated attention Transformer model trained using only tunnelling data. During the first 50 epochs, the loss values of both the training and validation sets decrease rapidly, suggesting that the model quickly learns effective features from the data at this stage. When the number of training epochs exceeds 100, the loss function tends to converge to its minimum value, indicating that 100 epochs are sufficient for completing the training process.

The proposed method demonstrates excellent performance in the PAA prediction task. The prediction results on the testing set are shown in Fig. 14. The overall histogram distributions of the predicted and monitored values are highly consistent, indicating high prediction accuracy. The evaluation metrics further confirm the superior performance of the model, with an R^2 value of 0.92, an RMSE of 0.69° , and an MAE of 0.39° , demonstrating high predictive accuracy. In the time-series comparison, the predicted curve agrees well with the measured curve in terms of the overall trend and overlaps closely with it at most time steps, indicating strong fitting capability. From time step 0 to approximately 10,000, the data points are relatively dense, and slight local deviations can be observed between the predicted and measured curves, indicating that the two curves do not fully overlap in this interval. However, considering the overall numerical range, the magnitudes of these deviations are generally small, and the prediction errors in this stage remain well controlled. In addition, during the rising segments between time steps 10,000–20,000 and 40,000–50,000, the predicted curve generally follows the trend of the measured curve, although noticeable deviations occur between a small number of data points and the corresponding predictions. These deviations constitute the main source of prediction error.

The prediction results of the gated attention Transformer model are shown in Fig. 15. In the scatter plot, the prediction values are closely distributed around the monitored values, with most

points located near both sides of the diagonal line, indicating high prediction accuracy. The evaluation metrics further demonstrate the model performance, with an R^2 value of 0.91, an RMSE of 0.74° , and an MAE of 0.40° . These results indicate that the model can explain most of the variance in the data, while maintaining the prediction error within a reasonable range. However, the time-series comparison reveals certain limitations of the model. Although the predicted curve generally follows the overall trend of the measured curve, evident deviations occur during several dynamically varying stages, and the corresponding error curve also exhibits pronounced fluctuations and peaks in these intervals. This suggests that the model remains limited in capturing rapid data variations or complex patterns, resulting in certain prediction biases.

Both the proposed DEAM method and the gated attention Transformer model achieve high prediction accuracy in the PAA prediction task. Compared with the gated attention Transformer model, DEAM increases the R^2 value from 0.91 to 0.92, corresponding to an improvement of 1.1 %. Meanwhile, the RMSE decreases from 0.74° to 0.69° , representing a reduction of approximately 6.8 %, and the MAE decreases from 0.40° to 0.39° , corresponding to a reduction of approximately 2.5 %. The time-series prediction results further indicate that, during stages where the gated attention Transformer model exhibits evident deviations, the prediction curve of DEAM aligns more closely with the measured curve, with substantially reduced error fluctuations. These results demonstrate the effectiveness of the decomposed framework adopted in this study.

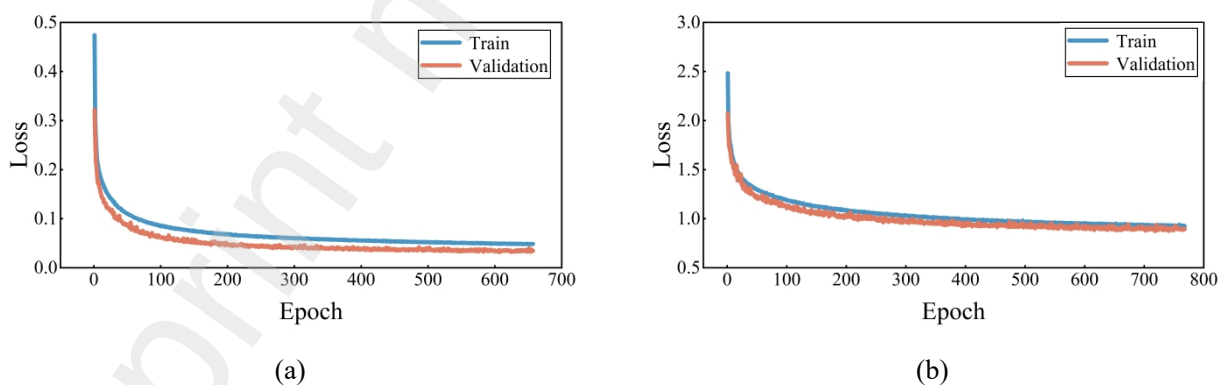


Fig. 12 Loss curves of the DEAM. (a) downtime attitude model, (b) excavation model

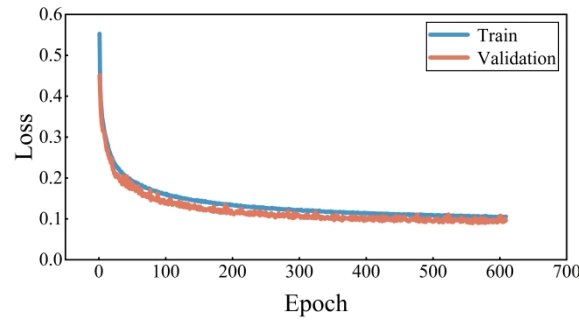


Fig. 13 Loss curves of the gated attention Transformer.

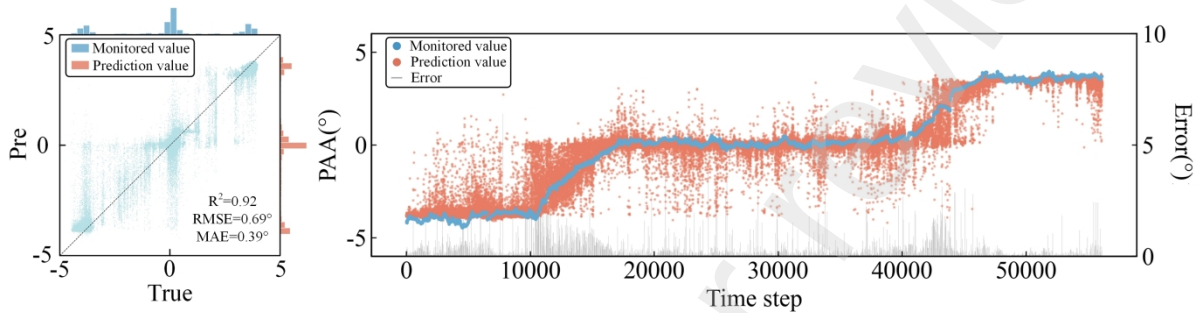


Fig. 14 Prediction results of the DEAM.

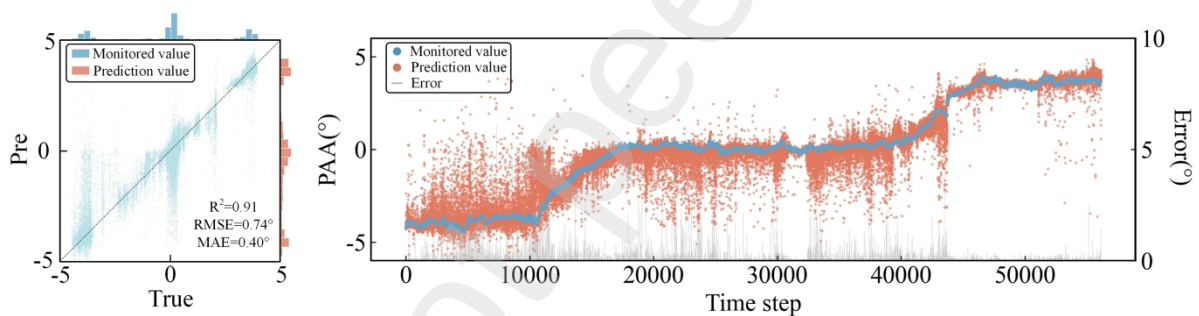


Fig. 15 Prediction results of the gated attention Transformer.

3.2.2 SHAP analysis results

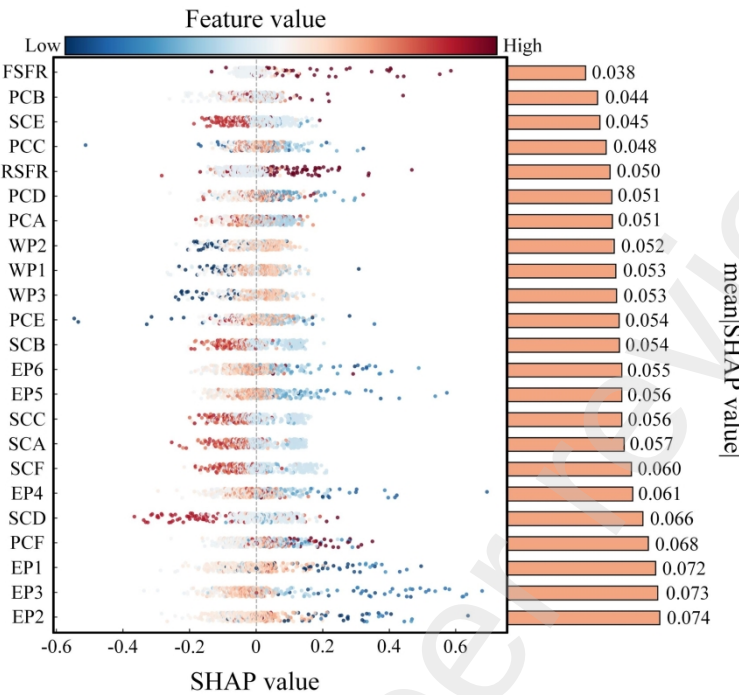
To identify the key input variables that exert significant influences on the model prediction results, the SHAP method was employed to quantify feature importance based on 500 randomly selected samples from the dataset. The visualization scheme provided by SHAP arranges the parameters along the central axis according to their importance and plots all samples in the corresponding coordinate space based on their SHAP values [31]. By quantifying the contribution of each input feature to the PAA prediction results, SHAP analysis reveals the key factors influencing shield machine attitude under different operating conditions and clarifies their directions of influence. This facilitates the identification of the dominant parameters governing PAA prediction under both static and dynamic conditions. The feature importance bar chart presents the mean absolute SHAP

value of each feature, where larger values indicate a more significant global influence on the model output. In the SHAP beeswarm plot, each point represents the SHAP value of a single sample. The color of each point denotes the magnitude of the feature value, where red indicates a relatively large feature value and blue indicates a relatively small feature value. The horizontal position of each point represents the sign and magnitude of the SHAP value, with positive values enhancing the prediction result whereas negative values suppress it.

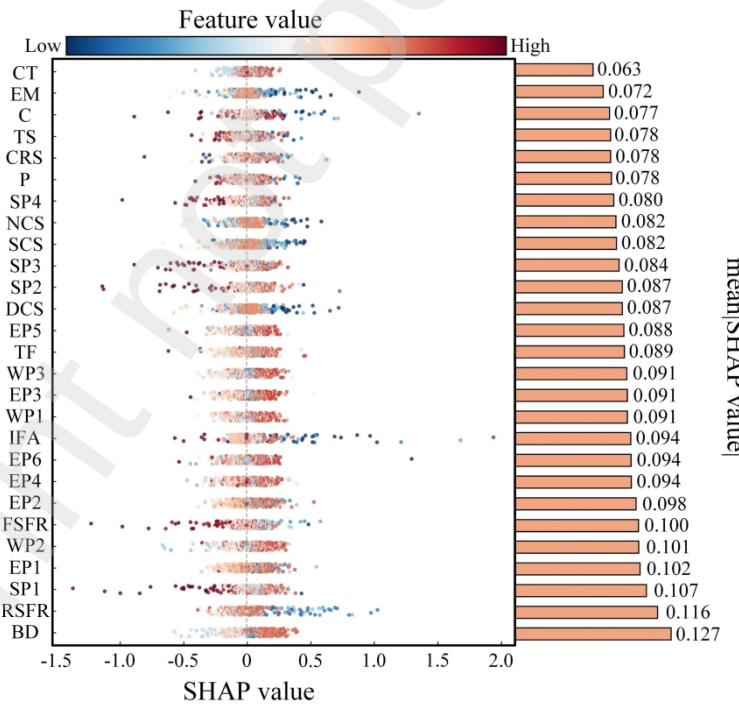
Fig. 16 presents the SHAP beeswarm plot and feature importance bar chart for the downtime attitude model. The three chamber pressure variables at the bottom exhibit the highest mean absolute SHAP values, indicating that they are the most influential factors affecting the output of the downtime attitude model. High values of chamber pressure are densely distributed in the negative SHAP value region, suggesting that higher chamber pressure significantly reduces the predicted output of the downtime prediction model. This indicates that sufficient excavation face support pressure helps stabilize the shield head and suppress upward pitching. High values of hydraulic cylinder pressure and cylinder stroke also tend to produce negative SHAP values, implying that the pressure of the main thrust cylinder groups is critical for stabilizing shield body attitude. The FSFR and RSFR rank lowest in feature importance, and their SHAP values are concentrated around zero. This suggests that slurry flow rate has a relatively limited influence on the PAA under downtime conditions.

Fig. 17 presents the SHAP beeswarm plot and feature importance bar chart for the excavation model. Burial depth and RSFR exhibit the highest mean absolute SHAP values, indicating that they are the primary factors influencing the output of the excavation model. High values of RSFR and low values of burial depth are concentrated in the positive SHAP value region. This indicates that a higher RSFR and shallower overburden depth significantly increase the model output, implying that excessive RSFR in shallow ground may readily induce upward pitching of the shield head. High shield tail grease pressure tends to produce negative SHAP values, suggesting that sufficient shield tail sealing pressure helps suppress attitude fluctuations. In contrast, high values of FSFR are associated with positive SHAP values, indicating that excessive slurry injection may also increase attitude instability. By comparison, tunnelling parameters and ground strength parameters, including CT, CRS, and P, generally show lower mean absolute SHAP values. This suggests that, in the

513 relatively homogeneous sandstone strata considered in this model, variations in tunnelling parameters
 514 and geotechnical strength have relatively limited effects on the dynamic fluctuations of the PAA.



515
 516 Fig. 16 The SHAP analysis results of the downtime attitude model.
 517



518
 519 Fig. 17 The SHAP analysis results of the excavation model.
 520

520 **4 Discussion**

521 To assess the accuracy of the proposed method, its performance was compared with that of six
 522 different methods trained using tunnelling data. Four of these methods are established techniques in

the field of time-series prediction: (1) convolutional neural network (CNN), a deep learning model widely used for time-series prediction, which captures local dependencies and spatial patterns in data through convolution operations; (2) recurrent neural network (RNN), a model designed for sequential modelling that captures temporal dynamics and contextual dependencies through recurrent structures; (3) MLP, a classical neural network architecture that performs nonlinear feature combination and mapping through fully connected layers in prediction tasks; and (4) Transformer, a self-attention-based model that explicitly captures correlations within sequences by modelling global dependencies and has shown outstanding performance in various prediction tasks in recent years. The fifth method replaces the downtime attitude model and excavation model in the proposed framework with encoders based on conventional self-attention, to verify the contribution of the gated attention to prediction accuracy. The sixth method removes geological parameters from the inputs of DEAM, to examine the influence of geological information on improving prediction accuracy.

4.1 Comparison with other models

4.1.1 CNN

CNN is a deep learning method capable of modelling complex nonlinear relationships and local temporal features in time-series data [32], and has therefore attracted extensive attention in prediction tasks. The model was implemented using the PyTorch framework. To improve model accuracy, normalized data were used as input, and the input dimensions were kept consistent with those of the proposed method.

The model adopted a one-dimensional convolutional architecture, with the convolution kernel size set to 3, ReLU selected as the activation function, the number of channels set to 32, and the learning rate and other parameters kept consistent with those of the proposed model. During training, MSE was used as the loss function, and the Adam optimizer was employed for parameter optimization.

4.1.2 RNN

RNN was originally proposed by Elman [33]. As a deep learning model specifically designed for sequential data, RNN can model dynamic dependencies and temporal patterns in time series through its internal recurrent connections, giving it unique advantages in sequence prediction tasks. The RNN method used for comparison consisted of a two-layer architecture, including one RNN

layer and one fully connected layer, with MSE adopted as the loss function. The input data of this model were processed in the same manner as those of the proposed model. The number of hidden units in the RNN layer was set to 5, and the fully connected layer was used to output the predicted attitude value. Model training was performed using the Adam optimizer, with the learning rate and other parameters kept consistent with those of the proposed model. Early stopping was applied during training.

4.1.3 MLP

MLP is a classical feedforward neural network that has been widely applied in regression prediction tasks owing to its strong nonlinear fitting capability and feature-space mapping ability [34]. The MLP method used for comparison consisted of a three-layer network architecture, including one fully connected hidden layer, one activation layer, and one output layer. MSE was adopted as the loss function, and the Adam optimizer was used for model training. The input data were standardized before training. Parameters such as the learning rate, number of training epochs, and batch size were kept consistent with those of the proposed model.

4.1.4 Transformer

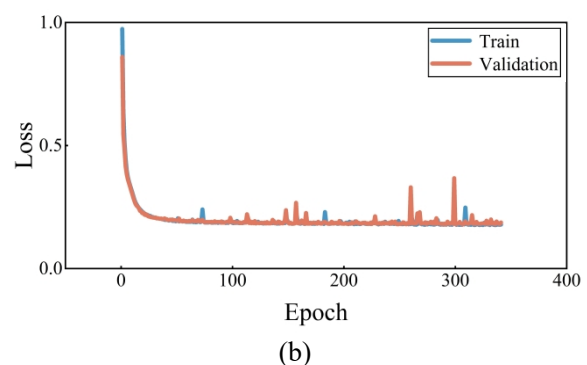
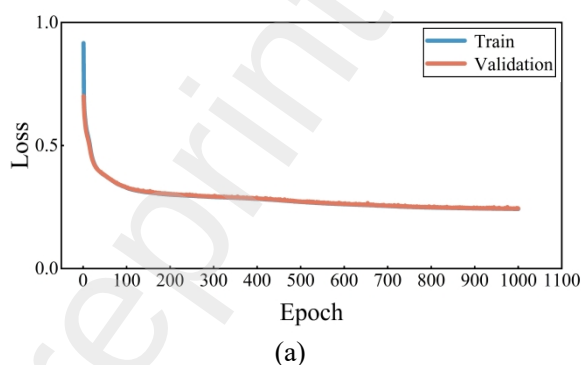
Transformer was proposed by Vaswani et al. [35] and has been widely applied in various fields, including time-series prediction [36]. As a deep learning method capable of modelling global dependencies and long-term sequential features, Transformer has demonstrated significant advantages in complex time-series prediction tasks. The Transformer method used for comparison consisted of a five-layer architecture, including one linear embedding layer, two Transformer encoder blocks, one global average pooling layer, and one regression output head. MSE was adopted as the loss function, and the Adam optimizer was used for model training. The input data were standardized based on the training set. Hyperparameters, including the learning rate, number of training epochs, batch size, and early stopping strategy, were kept consistent with those of the proposed model.

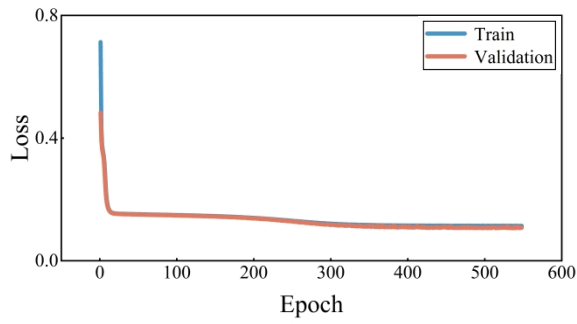
4.1.5 Comparative performance analysis

Fig. 18 and Fig. 19 present the training processes and prediction results of the models, respectively. The four models exhibit different levels of adaptability to the PAA prediction task. CNN can extract local temporal features, but relatively large deviations are observed between its prediction

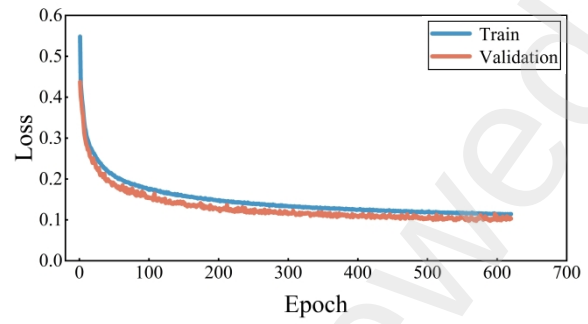
values and the monitored values. RNN can retain sequential information; however, large fluctuations tend to occur in intervals where the shield attitude changes sharply. MLP can generally track the main variation trend of the monitored values, but its prediction error increases during rapid attitude changes. Transformer achieves the best performance among the baseline models, because the self-attention mechanism enables it to model global dependencies. Nevertheless, noticeable local errors still exist in some intervals with complex variations. These results indicate that a single model has difficulty fully characterizing shield attitude variations in practical engineering scenarios.

Fig. 20 presents the performance metrics of different models in the PAA prediction task. The R^2 values of CNN, RNN, MLP, Transformer, and DEAM increase progressively, reaching 0.76, 0.82, 0.89, 0.90, and 0.92, respectively. CNN has the largest prediction errors, with an RMSE of 1.22° and an MAE of 0.95° . The errors of RNN and MLP are reduced to some extent, with RMSE values of 1.04° and 0.81° and MAE values of 0.66° and 0.63° , respectively. Transformer further reduces the RMSE and MAE to 0.78° and 0.42° , indicating that the self-attention mechanism can capture long-range temporal dependencies. Finally, the proposed DEAM achieves the lowest prediction errors, reducing RMSE and MAE to 0.69° and 0.39° , respectively. Compared with Transformer, DEAM reduces the RMSE and MAE by approximately 11.54 % and 7.14 %, respectively. Compared with CNN, the RMSE and MAE are reduced by approximately 43.44 % and 58.95 %, respectively. These results demonstrate that DEAM outperforms the other comparative models in terms of both prediction accuracy and error control, achieving higher fitting accuracy and lower prediction errors, thereby validating its effectiveness in shield attitude prediction tasks.



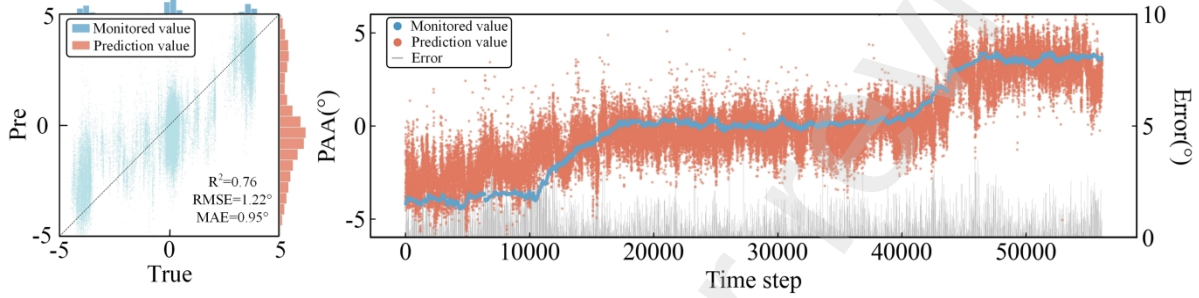


(c)

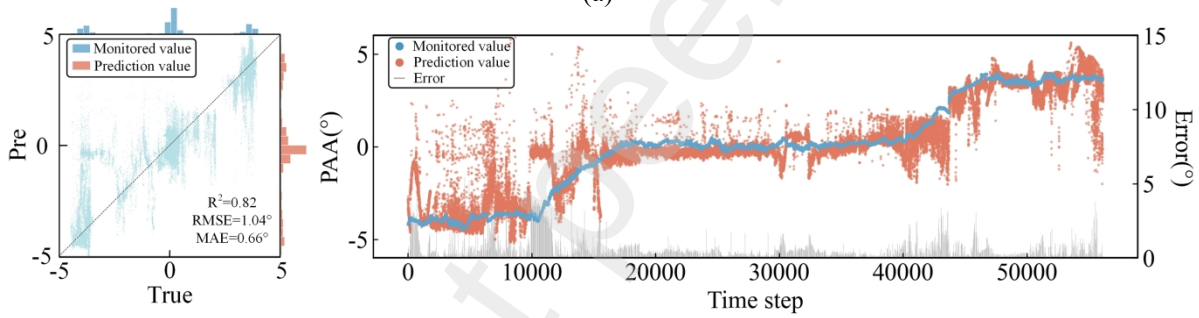


(d)

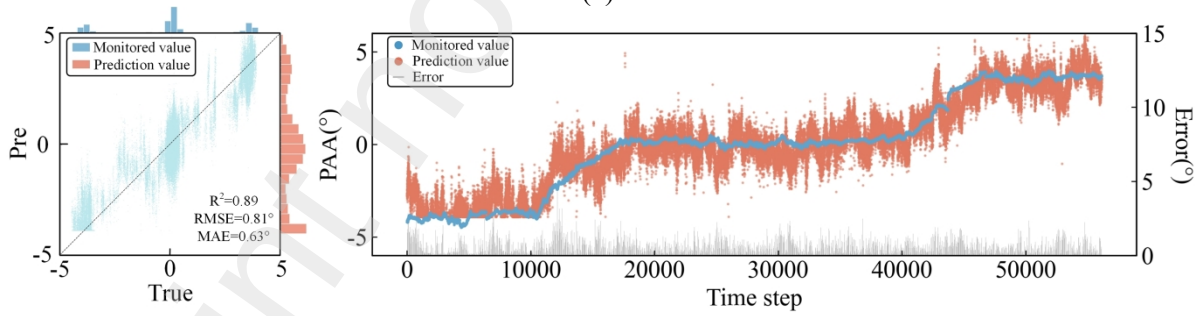
Fig. 18 Loss curves of models. (a) CNN, (b) RNN, (c) MLP, (d) Transformer.



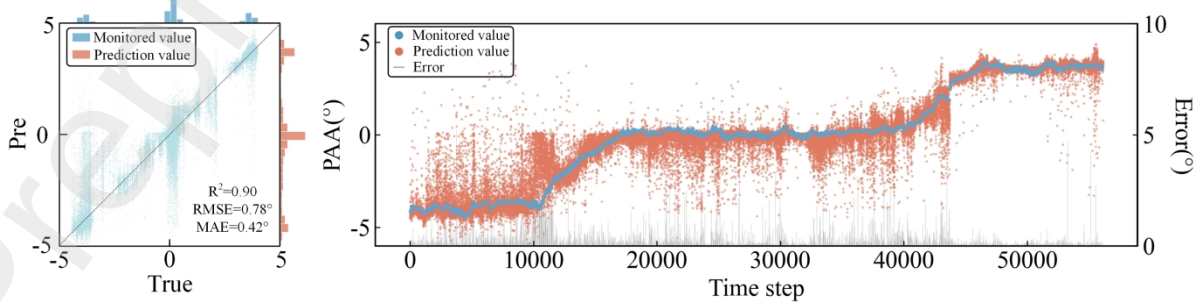
(a)



(b)



(c)



(d)

Fig. 19 Prediction results of models. (a) CNN, (b) RNN, (c) MLP, (d) Transformer.

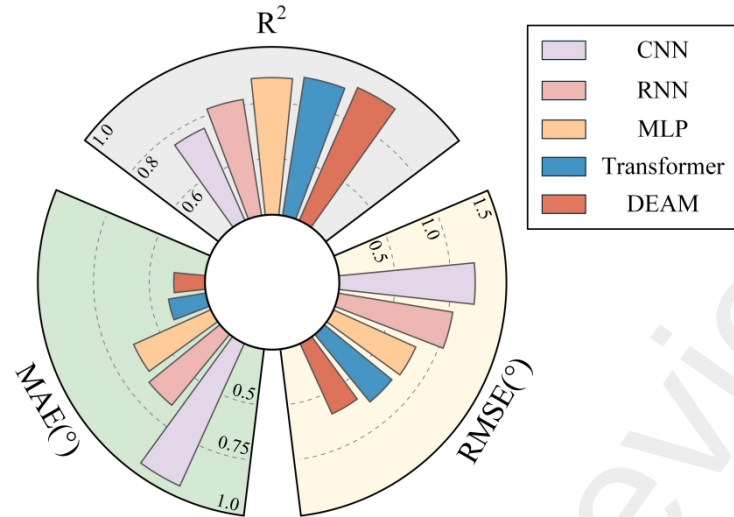


Fig. 20 Quantitative performance comparison of different models

4.2 The effectiveness of gated attention

To verify the contribution of the gated attention to improving prediction accuracy, an ablation experiment was designed by replacing the gated attention module in the proposed model with the original self-attention module without gating. The remaining architecture and hyperparameters were kept consistent with those used in the main experiment. The model was trained under the same conditions, and the training process and prediction results are shown in Fig. 21 and Fig. 22.

Compared with the proposed DEAM method, the model without the gated attention shows a rapid decrease in the training loss of the downtime attitude model at the initial stage, whereas the excavation model converges more slowly. After convergence, the loss curves of the model without gated attention still exhibit slight fluctuations. Although similar fluctuations are also observed in the later convergence stage of DEAM, their amplitudes are smaller, indicating that the gated attention improves model stability. This improvement can be attributed to the sparsity-oriented modulation of attention weights introduced by the gating mechanism, which suppresses redundant information flow and enhances model stability. The introduction of gated attention increases R^2 by approximately 1.1 %, reduces RMSE by 8.7 %, and reduces MAE by 7.7 %. Moreover, during dynamically varying stages, the prediction values of DEAM align more closely with the monitored values, and the corresponding error curve is smaller. This indicates that gated attention reduces prediction errors under complex construction conditions by enhancing the model's ability to capture critical features, further confirming the effectiveness of the gated attention.

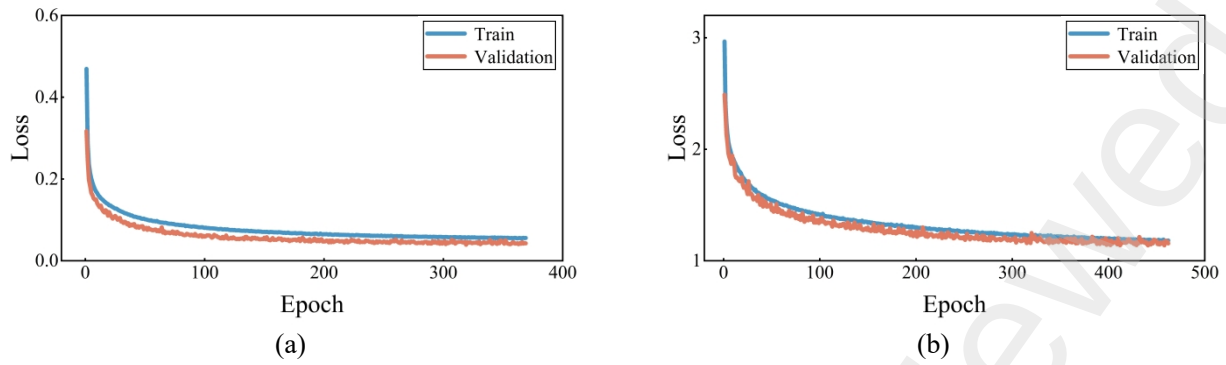


Fig. 21 Loss curves of DEAM without gated attention. (a) downtime attitude model, (b) excavation model

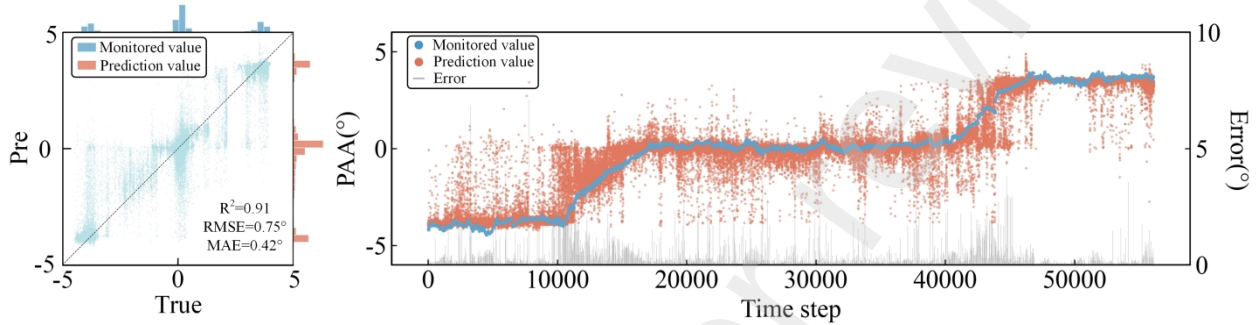


Fig. 22 Prediction results of DEAM without gated attention.

4.3 Proposed model on data without geological parameters

To verify the contribution of geological parameters to improving prediction accuracy, an ablation experiment was designed by removing geological parameters from the input variables of the proposed model. The model architecture and hyperparameters were kept consistent with those used in the main experiment. The model was trained under the same conditions, and the training process and prediction results are shown in Fig. 23 and Fig. 24.

Compared with the proposed DEAM method, the model without geological parameters shows unchanged training loss for the downtime attitude model. In contrast, the excavation model exhibits higher loss values during the initial training stage and a higher final convergence value. Although the model without geological parameters converges faster, the complete DEAM model shows smaller fluctuations in the loss curve. This indicates that geological parameters provide critical information for the model, and their absence reduces the stability of the learning process and limits sufficient convergence. Removing geological parameters leads to a significant degradation in model performance, with R^2 decreasing by 3.4 %, RMSE increasing by 17.4 %, and MAE increasing by 20.5 %. During dynamically varying stages, the predictions of DEAM align more closely with the monitored values, with substantially smaller errors. These results demonstrate that geological

parameters characterize engineering geological conditions, enabling the model to more accurately predict shield attitude variations under complex construction conditions and thereby improving prediction accuracy.

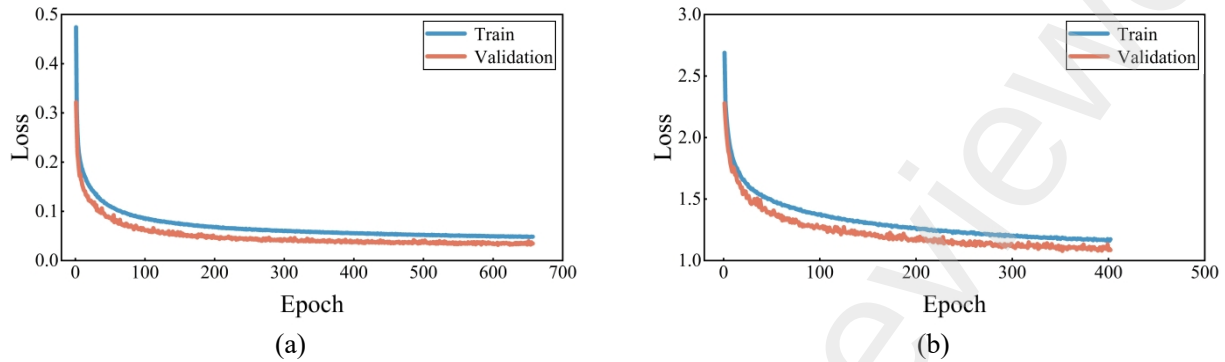


Fig. 23 Loss curves of DEAM without geological parameters. (a) downtime attitude model, (b) excavation model

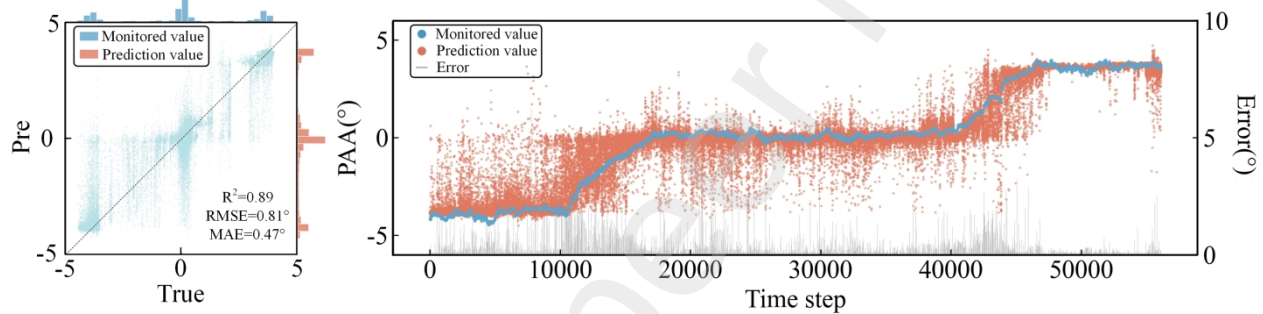


Fig. 24 Prediction results of DEAM without geological parameters.

5 Conclusion

This study proposed an intelligent prediction method for key shield attitude parameters based on a physical model of static soil-shield interaction, termed DEAM. Downtime data and tunnelling data were used as the inputs of the downtime attitude model and excavation model, respectively, and gated attention was incorporated to improve the accuracy and stability of PAA prediction. The proposed method was empirically validated using data from the eastbound line of the Guangzhou Haizhu Bay Tunnel project and compared with various deep learning models. Based on the experimental results and discussion, the following conclusions can be drawn:

(1) The DEAM method characterizes the attitude response mechanisms of the shield machine under downtime equilibrium and dynamic tunnelling conditions through a static-dynamic decomposition framework. The downtime attitude model establishes a quasi-static mapping between loading conditions and attitude using parameters such as hydraulic cylinder thrust and chamber pressure, whereas the excavation model corrects the predicted PAA by capturing dynamic features

656 from tunnelling data. This decomposition strategy significantly enhances the model's capability to
657 represent nonlinear relationships under complex construction conditions.

658 (2) SHAP-based feature importance analysis indicates that chamber pressure and hydraulic
659 cylinder parameters are the dominant factors governing PAA prediction in the downtime attitude
660 model, whereas burial depth and return slurry flow rate exert significant influences in the excavation
661 model. These findings reveal the distinct roles of key parameters under downtime and tunnelling
662 states, providing useful guidance for optimizing attitude control parameters in engineering practice.

663 (3) Compared with the baseline models, DEAM achieves the best performance in the PAA
664 prediction task, with an R^2 of 0.92 and RMSE and MAE values reduced to 0.69° and 0.39° ,
665 respectively. Compared with Transformer, DEAM reduces the RMSE and MAE by approximately
666 11.54 % and 7.14 %, respectively; compared with CNN, the corresponding reductions reach
667 approximately 43.44 % and 58.95 %. In addition, the ablation experiments show that gated attention
668 further improves model performance, increasing R^2 by approximately 1.1 % and reducing RMSE and
669 MAE by approximately 8.7 % and 7.7 %, respectively. After removing geological parameters, the
670 training stability of the model decreases, with R^2 reduced by approximately 3.4 % and RMSE and
671 MAE increased by approximately 17.4 % and 20.5 %, respectively. These results validate the
672 effectiveness and reliability of DEAM for shield attitude prediction under complex construction
673 conditions.

674 Acknowledgements

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